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A Hybrid Reinforcement Learning and Harris Hawks Optimization Framework for Multi-Objective Resource Management in IIoT

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ABSTRACT

The Industrial Internet of Things (IIoT) is revolutionizing industrial processes by interconnecting sensors, machines, and computing resources to enable intelligent automation, real-time monitoring, and predictive maintenance. Despite its potential, efficient allocation and management of resources in IIoT networks remain a significant challenge due to the dynamic, heterogeneous, and large-scale nature of industrial environments. Traditional static or heuristic resource allocation methods often fail to satisfy conflicting objectives such as minimizing latency, maximizing throughput, ensuring energy efficiency, and maintaining system reliability. To address these challenges, this paper presents the design and development of an AI-based multi-objective resource management system for IIoT systems. The system integrates reinforcement learning (RL) for adaptive, real-time decision-making with Harris Hawks Optimization (HHO) for multi-objective optimization, balancing competing performance metrics across sensing, communication, computing, and control layers. The RL agent predicts system states and dynamically adjusts resource allocation, while HHO fine-tunes the solutions to achieve optimal performance under multiple constraints. Simulation studies demonstrate that the proposed system outperforms conventional static and heuristic approaches, achieving higher efficiency, faster adaptability, and improved scalability. By intelligently managing resources and optimizing system behavior, this system enhances overall operational reliability and Quality-of-Service in IIoT networks. The results indicate that combining AI-driven decision-making with advanced nature-inspired optimization provides a robust and efficient solution for next-generation industrial applications, paving the way for more intelligent, resilient, and resource-efficient IIoT systems.

Please note: Abbreviations should be introduced at the first mention in the main text – no abbreviations lists. Suggested structure of main text (not enforced) is provided below.

1 INTRODUCTION

The Industrial Internet of Things (IIoT) is transforming modern industries by connecting machines, sensors, and computing resources to create intelligent, automated, and interconnected networks¹. These systems enable real-time monitoring, predictive maintenance, and efficient industrial operations, contributing to increased productivity, reduced downtime, and better decision-making. Despite these advancements, managing resources in IIoT remains a significant challenge². Industrial networks are inherently dynamic and heterogeneous, where devices range from low-power sensors and actuators to high-performance edge servers, each with distinct computational, communication, and energy capabilities. The resource demands of IIoT applications fluctuate continuously due to variations in sensing data rates, control loop deadlines, and task offloading requirements, making static or heuristic allocation strategies ineffective. Additionally, conflicting objectives such as minimizing latency while reducing energy consumption and ensuring fairness among devices introduce a complex multi-objective optimization problem that cannot be efficiently solved with conventional methods. Without an adaptive and intelligent mechanism to balance these factors, IIoT systems risk underutilization of resources, increased constraint violations, and degraded quality of service. Efficient allocation of computing power, communication bandwidth, and control capacity is critical to meet stringent requirements for latency, throughput, energy efficiency, and reliability. Traditional resource allocation strategies that rely on static rules or heuristic

approaches often fail to adapt to these fluctuating conditions, resulting in suboptimal performance and limited scalability.

Addressing these challenges requires intelligent and adaptive methods that can handle multiple objectives simultaneously. This study proposes the design and development of an AI-based multi-objective resource management system for IIoTs. The system integrates reinforcement learning (RL) for adaptive, real-time decision-making with Harris Hawks Optimization (HHO), a nature-inspired algorithm, to efficiently handle multi-objective optimization. Reinforcement learning enables the system to anticipate network conditions and make dynamic allocation decisions, while HHO fine-tunes these decisions to achieve optimal performance across multiple criteria, including latency minimization, throughput maximization, energy efficiency, and reliability. This hybrid approach ensures that the system can adapt to real-time changes while maintaining globally optimized resource allocations.

Through extensive simulations, the proposed system demonstrates substantial improvements over conventional static and heuristic methods. It achieves faster adaptability, higher energy efficiency, lower latency, and improved throughput, highlighting its practical relevance for industrial deployments. By combining learning and optimization, the system provides a scalable, robust, and intelligent solution for IIoT networks, enabling more resilient and efficient industrial operations.

1.1 MAJOR CONTRIBUTIONS

This paper makes several significant contributions to the field of resource management in IIoT systems.

- A comprehensive system model is developed for IIoT environments that jointly captures task arrivals, computational resources, energy consumption, delay constraints, and reliability requirements, enabling a realistic representation of industrial resource management scenarios.
- A hybrid resource management system is proposed that integrates reinforcement learning with Harris Hawks Optimization, where reinforcement learning enables adaptive policy learning from system feedback and HHO refines resource allocation decisions through efficient global optimization.
- A multi-objective optimization formulation is designed to simultaneously optimize throughput, energy consumption, latency, and resource utilization while explicitly considering system constraints, thereby reflecting practical industrial requirements.
- The proposed system incorporates constraint-aware optimization to minimize violation rates and improve operational robustness, ensuring feasible and stable performance under varying traffic conditions.
- Extensive simulations are conducted under different arrival rates to evaluate the effectiveness of the proposed approach, and performance is compared against advanced benchmark algorithms including GA, WOA, RL-only, and RL+PSO.
- Detailed performance analysis using multiple metrics such as throughput, average latency, energy consumption, resource utilization, system utility, reliability, convergence behavior, and joint performance index demonstrates the superiority and scalability of the proposed system in dynamic IIoT environments.

The rest of the paper is organized as follows. Section II presents the related works on resource allocation techniques in IIoT, highlighting their limitations. Section III introduces the network model and key definitions. Section IV formulates the resource allocation problem using linear programming. Section V describes the prerequisites of the proposed system, including an overview of reinforcement learning and Harris Hawks Optimization. Section VI details the design of the proposed hybrid RL-HHO system, including workflow, algorithms, and utility formulation. Section VII reports the simulation setup, performance metrics, and comparative evaluation with baseline methods. Section VIII concludes the paper with a summary of contributions and outlines possible future research directions.

2 RELATED WORKS

The existing literature on resource management in the IIoT spans a variety of methods, ranging from classical scheduling algorithms to advanced machine learning approaches³⁻⁵.

Early techniques such as round robin (RR) is one of the earliest task allocation techniques applied to IIoT environments. It ensures fairness by distributing tasks cyclically across resources⁶. However, RR fails to consider workload heterogeneity, often resulting in high latency and poor energy efficiency in dynamic industrial networks. The genetic algorithm (GA) has been applied for resource allocation in IIoT due to its ability to explore large search spaces. It adapts solutions iteratively through crossover and mutation, achieving better utilization compared to static methods⁷. Yet, GA suffers from slow convergence and can be computationally expensive under real-time constraints. In⁸, the authors have proposed a whale optimization algorithm (WOA) for balancing the load and energy efficiency. However, it primarily relies on static server configurations and

Table 1. List of Abbreviations

Abbreviation	Full Form
ACO	Ant Colony Optimization
AI	Artificial Intelligence
CAD	Computer-Aided Design
CPU	Central Processing Unit
DQN	Deep Q-Network
GA	Genetic Algorithm
HHO	Harris Hawks Optimization
IIoT	Industrial Internet of Things
IoT	Internet of Things
MDP	Markov Decision Process
PSO	Particle Swarm Optimization
QoS	Quality-of-Service
RL	Reinforcement Learning
RR	Round Robin
WOA	Whale Optimization Algorithm

limited scalability scenarios. Moreover, the study lacks a unified multi-objective formulation that jointly considers latency, reliability, and constraint violations. The RL-based methods enable adaptive decision-making by learning optimal allocation policies from interaction with the environment⁹. They outperform heuristic approaches in throughput and latency optimization. However, RL often converges slowly and may fall into local optima, especially under highly variable workloads. The deep Q-network (DQN) for IIoT Scheduling extends RL with neural function approximation, improving scalability to large IIoT state spaces¹⁰. It significantly reduces decision time and improves system utility. Despite its benefits, it requires large training datasets and can be computationally heavy for resource-constrained devices. Particle swarm optimization (PSO)-based resource management has been explored for multi-objective optimization in IIoT networks, balancing energy efficiency and throughput¹¹. Its parallel search mechanism allows faster convergence than GA. However, PSO tends to stagnate in high-dimensional spaces and struggles with rapidly changing environments. Fuzzy Logic-based task allocation inference systems have been applied for decision-making in heterogeneous IIoT networks¹². They provide interpretable and flexible solutions for uncertain environments. Nonetheless, their performance depends heavily on expert-defined rules, limiting adaptability in large-scale dynamic deployments. Several works have explored task offloading to edge servers for latency reduction¹³. These methods improve real-time responsiveness compared to cloud-only solutions. Still, they often ignore global optimization, leading to imbalanced resource utilization when edge resources are limited. Markov decision process (MDP)-based resource allocation formulations have been applied to capture the stochastic behavior of IIoT workloads¹⁴. They provide mathematically sound solutions for sequential decision-making. Yet, their scalability is limited in large industrial networks due to state explosion. The ant colony optimization (ACO) for IIoT resource scheduling inspired by swarm intelligence that is used for dynamic path and resource optimization in IIoT¹⁵. It shows better adaptability to fluctuating workloads compared to static methods. However, ACO has slow convergence and requires significant fine-tuning of parameters. Recent works propose combining machine learning models with heuristic optimizers to balance adaptability and computational efficiency¹⁶. These hybrid models demonstrate better performance in multi-objective optimization. However, most studies remain simulation-based, lacking validation in real industrial testbeds.

From the review of existing approaches, it is evident that no single method fully addresses the challenges of IIoT resource management. These gaps highlight the need for a hybrid approach that combines the adaptability of reinforcement learning with the global search capabilities of metaheuristic optimization. The proposed RL+HHO system is designed to fill this gap, offering a scalable and intelligent solution to meet the multi-objective demands of IIoT networks.

3 NETWORK MODEL

We consider an IIoT network consisting of heterogeneous devices interconnected via wireless links. The network integrates four layers: sensing, communication, computing, and control. Let the set of devices be $\mathcal{D} = \{1, 2, \dots, D\}$ and the set of tasks to be executed be $\mathcal{T} = \{1, 2, \dots, T\}$. Communication links between devices are assumed to have finite bandwidth and are denoted by L_{ij} , representing the link between devices i and j .

3.1 NETWORK COMPONENTS AND RESOURCES

Each device $d \in \mathcal{D}$ is characterized by the following resources:

- *Sensing Resource* (S_d): The capability of device d to collect data from the environment.
- *Communication Resource* (C_d): Bandwidth available for transmitting task-related data.
- *Computing Resource* (P_d): CPU or processing capacity for executing tasks.
- *Control Resource* (K_d): Ability to perform actuation or control operations within latency requirements.

Each task $t \in \mathcal{T}$ is defined by:

- *Workload* (w_t): Total computational requirement in CPU cycles.
- *Data Size* (d_t): Volume of data to be transmitted or processed.
- *Latency Requirement* ($L_t^{\max}$): Maximum allowable time for task completion.

The resource allocation vector for task t at device d is given by:

$$\mathbf{x}_t = [s_t, c_t, p_t, k_t], \quad (1)$$

where s_t, c_t, p_t, k_t are allocated sensing, communication, computing, and control resources, respectively.

3.2 Network Utility Function

The network aims to maximize a weighted combination of throughput, latency satisfaction, and energy efficiency:

$$\max_{\mathbf{x}} U(\mathbf{x}) = \alpha \sum_{t \in \mathcal{T}} T_t - \beta \sum_{t \in \mathcal{T}} L_t - \gamma \sum_{d \in \mathcal{D}} E_d \quad (2)$$

where:

$$E_d = \eta_s s_t + \eta_c c_t + \eta_p p_t + \eta_k k_t, \quad (3)$$

$$T_t = \phi_c c_t + \phi_p p_t, \quad (4)$$

$$L_t = \lambda_s s_t + \lambda_c c_t + \lambda_p p_t + \lambda_k k_t. \quad (5)$$

Here, α, β, γ are weighting factors, η, ϕ , and λ are coefficients capturing energy, throughput, and latency contributions of each resource type.

4 PROBLEM FORMULATION

The resource allocation problem is formulated as a linear programming problem:

$$\max_{\mathbf{x}} U(\mathbf{x}) \quad (6)$$

$$\text{subject to: } L_t \leq L_t^{\max}, \quad \forall t \in \mathcal{T} \quad (7)$$

$$\sum_{t \in \mathcal{T}} E_d \leq E_d^{\max}, \quad \forall d \in \mathcal{D} \quad (8)$$

$$\sum_{t \in \mathcal{T}} c_t \leq C_d^{\max}, \quad \forall d \in \mathcal{D} \quad (9)$$

$$\sum_{t \in \mathcal{T}} p_t \leq P_d^{\max}, \quad \forall d \in \mathcal{D} \quad (10)$$

$$s_t, c_t, p_t, k_t \geq 0, \quad \forall t \in \mathcal{T}. \quad (11)$$

The Eq. (7) ensures latency constraint i.e., the task completion within maximum allowed time. Then, Eq. (8) denotes the energy constraint that limits the energy consumption of all devices. The bandwidth constraint in eq. (9) Prevents exceeding available communication resources. Further, the eq. (10) ensures computing constraint i.e., the CPU allocations do not exceed device limits. Similarly, the non-negativity constraint Guarantees feasible, positive resource allocations in eq. (11).

5 PREREQUISITES REQUISITES OF THE PROPOSED SYSTEM

5.1 Overview of Reinforcement Learning

The RL is a machine learning paradigm where an agent learns to make sequential decisions by interacting with an environment to maximize cumulative reward. In dynamic IIoT networks, task arrivals, device states, and resource availability change continuously. RL is particularly suitable for adaptive resource allocation in such networks. Let the network state at time t be represented as:

$$S_t = [U_d, Q_t, E_d, L_t], \quad \forall d \in \mathcal{D}, t \in \mathcal{T} \quad (12)$$

where, U_d is device utilization, Q_t is task backlog, E_d is remaining energy, and L_t is task latency.

The action of the agent corresponds to the resource allocation vector:

$$A_t = \mathbf{x}_t = [s_t, c_t, p_t, k_t], \quad s_t, c_t, p_t, k_t \geq 0 \quad (13)$$

The reward function reflects system performance by combining throughput, latency, and energy efficiency:

$$R_t = \alpha \sum_{t \in \mathcal{T}} T_t - \beta \sum_{t \in \mathcal{T}} L_t - \gamma \sum_{d \in \mathcal{D}} E_d \quad (14)$$

The Q-learning update rule for policy improvement is:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \eta \times [R_t + \gamma \times \max_{A_{t+1}} Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)] \quad (15)$$

$$\quad (16)$$

where, η is the learning rate and γ the discount factor. Deep Q-Networks (DQN) approximate $Q(S_t, A_t)$ for large state-action spaces.

5.1.1 Use of RL in the Proposed System

In our method, RL serves as a coarse-grained allocator. At each time step, the agent observes the network state S_t and proposes an initial resource allocation:

$$\mathbf{x}_t^{\text{RL}} = [s_t^{\text{RL}}, c_t^{\text{RL}}, p_t^{\text{RL}}, k_t^{\text{RL}}] \quad (17)$$

These allocations approximately satisfy constraints:

$$\sum_{t \in \mathcal{T}} c_t^{\text{RL}} \leq C_d^{\text{max}}, \quad \sum_{t \in \mathcal{T}} p_t^{\text{RL}} \leq P_d^{\text{max}}, \quad L_t^{\text{RL}} \leq L_t^{\text{max}} \quad (18)$$

The reward for this allocation is:

$$R_t = \alpha \sum T_t(\mathbf{x}_t^{\text{RL}}) - \beta \sum L_t(\mathbf{x}_t^{\text{RL}}) - \gamma \sum E_d(\mathbf{x}_t^{\text{RL}}) \quad (19)$$

The agent updates its policy using:

$$\pi(S_t) = \arg \max_{A_t} Q(S_t, A_t) \quad (20)$$

The RL-generated allocation $\mathbf{x}_t^{\text{RL}}$ acts as a baseline for the HHO algorithm, which refines the allocation to produce a final solution:

$$\mathbf{x}_t^{\text{final}} = \text{HHO}(\mathbf{x}_t^{\text{RL}}, S_t) \quad (21)$$

This integration allows the system to adapt to dynamic network conditions while ensuring constraint compliance, high throughput, low latency, and energy efficiency in real time.

5.2 OVERVIEW OF HARRIS HAWKS OPTIMIZATION

The HHO is a nature-inspired metaheuristic algorithm that mimics the cooperative hunting behavior of Harris hawks. It is particularly effective for multi-objective and high-dimensional optimization problems, making it suitable for refining resource allocations in IIoT networks. Let the resource allocation solution at iteration t be denoted as:

$$\mathbf{X}_t = [s_t, c_t, p_t, k_t], \quad \forall t \in \mathcal{T} \quad (22)$$

The HHO operates in two main phases: *exploration* and *exploitation*. During exploration, hawks search globally for promising solutions using:

$$\mathbf{X}_{t+1} = \mathbf{X}_r - r_1 |\mathbf{X}_r - 2r_2 \mathbf{X}_t| \quad (23)$$

where, $\mathbf{X}_r$ is a randomly selected hawk position, and $r_1, r_2 \in [0, 1]$ are random numbers.

During exploitation, hawks converge on the prey (optimal solution) using soft besiege or hard besiege strategies, depending on the escaping energy E of the prey:

$$\mathbf{X}_{t+1} = \mathbf{X}_{\text{prey}} - E |J \mathbf{X}_{\text{prey}} - \mathbf{X}_t| \quad (24)$$

where, J is a random number in $[0, 2]$, controlling jump strength, and E represents the prey's energy decreasing over iterations. The fitness function guiding the hawks is the system utility:

$$U(\mathbf{X}_t) = \alpha \sum T_t - \beta \sum L_t - \gamma \sum E_d \quad (25)$$

which ensures the algorithm seeks allocations maximizing throughput, reducing latency, and minimizing energy consumption.

5.2.1 Use of HHO in the Proposed System

In the proposed system, HHO refines the initial allocations generated by the RL agent. The RL output $\mathbf{x}_t^{\text{RL}}$ serves as the starting population for HHO:

$$\mathbf{X}_0 = \mathbf{x}_t^{\text{RL}} \quad (26)$$

The HHO algorithm iteratively updates each solution $\mathbf{X}_t$ to improve the system utility while satisfying constraints:

$$\begin{aligned} L_t(\mathbf{X}_t) &\leq L_t^{\max}, \quad \forall t \in \mathcal{T} \\ \sum_{t \in \mathcal{T}} c_t(\mathbf{X}_t) &\leq C_d^{\max}, \quad \forall d \in \mathcal{D} \\ \sum_{t \in \mathcal{T}} p_t(\mathbf{X}_t) &\leq P_d^{\max}, \quad \forall d \in \mathcal{D} \\ s_t, c_t, p_t, k_t &\geq 0 \end{aligned} \quad (27)$$

At each iteration, the positions of hawks are updated according to exploration or exploitation strategies, and the fitness of each candidate is evaluated using $U(\mathbf{X}_t)$. The process continues until convergence or the maximum number of iterations is reached, producing a final refined allocation:

$$\mathbf{x}_t^{\text{final}} = \arg \max U(\mathbf{X}_t) \quad (28)$$

By using HHO, the system achieves fine-grained optimization of RL-provided allocations, ensuring that all constraints are strictly met while maximizing throughput, minimizing latency, and improving energy efficiency. The hybrid RL-HHO approach allows the system to be adaptive, efficient, and robust, handling dynamic network conditions in real time.

6 PROPOSED RESOURCE MANAGEMENT SYSTEM

The proposed method aims to achieve adaptive and efficient resource allocation in IIoT networks. By integrating a RL agent with the HHO algorithm, the system ensures that sensing, communication, computing, and control resources are allocated optimally while satisfying latency, energy, and bandwidth constraints. The system operates in two layers:

1. *RL Layer (Coarse Allocation)*: Generates an initial allocation for tasks based on the current network state.
2. *HHO Layer (Fine-Tuning)*: Refines RL-generated allocations to meet all constraints and maximize system utility.

This hybrid approach leverages RL's adaptive decision-making and HHO's efficient global and local search capabilities to achieve near-optimal allocations in real time.

6.1 SYSTEM WORKFLOW

The resource allocation process follows these steps:

1. *State Observation*: At each discrete time step t , the RL agent observes the network state:

$$S_t = [U_d, Q_t, E_d, L_t], \quad \forall d \in \mathcal{D}, t \in \mathcal{T} \quad (29)$$

2. *RL-Based Allocation*: The agent proposes an initial allocation vector:

$$\mathbf{x}_t^{\text{RL}} = [s_t^{\text{RL}}, c_t^{\text{RL}}, p_t^{\text{RL}}, k_t^{\text{RL}}] \quad (30)$$

This allocation satisfies approximate constraints:

$$\begin{aligned} \sum_{t \in \mathcal{T}} c_t^{\text{RL}} &\leq C_d^{\text{max}}, \quad \sum_{t \in \mathcal{T}} p_t^{\text{RL}} \leq P_d^{\text{max}}, \\ L_t^{\text{RL}} &\leq L_t^{\text{max}}, \quad s_t, c_t, p_t, k_t \geq 0 \end{aligned} \quad (31)$$

3. *Reward Computation*: The system utility is calculated for the RL allocation:

$$R_t = \alpha \sum T_t(\mathbf{x}_t^{\text{RL}}) - \beta \sum L_t(\mathbf{x}_t^{\text{RL}}) - \gamma \sum E_d(\mathbf{x}_t^{\text{RL}}) \quad (32)$$

4. *HHO-Based Refinement*: The RL allocation $\mathbf{x}_t^{\text{RL}}$ serves as the initial population for HHO:

$$\mathbf{X}_0 = \mathbf{x}_t^{\text{RL}} \quad (33)$$

The HHO iteratively updates the positions of hawks according to exploration and exploitation strategies:

$$\begin{aligned} \mathbf{X}_{t+1} &= \mathbf{X}_r - r_1 |\mathbf{X}_r - 2r_2 \mathbf{X}_t|, \quad (\text{Exploration}) \\ \mathbf{X}_{t+1} &= \mathbf{X}_{\text{prey}} - E |J \mathbf{X}_{\text{prey}} - \mathbf{X}_t|, \quad (\text{Exploitation}) \end{aligned} \quad (34)$$

5. *Constraint Verification*: Each candidate solution is checked against:

$$\begin{aligned} L_t(\mathbf{X}_t) &\leq L_t^{\text{max}}, \quad \sum_{t \in \mathcal{T}} c_t(\mathbf{X}_t) \leq C_d^{\text{max}}, \\ \sum_{t \in \mathcal{T}} p_t(\mathbf{X}_t) &\leq P_d^{\text{max}}, \quad s_t, c_t, p_t, k_t \geq 0 \end{aligned} \quad (35)$$

6. *Final Allocation*: The refined allocation with the highest system utility is applied:

$$\mathbf{x}_t^{\text{final}} = \arg \max U(\mathbf{X}_t) \quad (36)$$

7. *Policy Update*: RL receives feedback from the refined allocation to improve its future decisions.

Algorithm 1 RL-Based Coarse Resource Allocation

Require: Network state $S_t = [U_d, Q_t, E_d, L_t]$, RL agent parameters, max episodes N

Ensure: Initial resource allocation $\mathbf{x}_t^{\text{RL}}$

- 1: **for** episode = 1 to N **do**
- 2: Observe current state S_t
- 3: Select action A_t using policy $\pi(S_t)$ or ϵ -greedy
- 4: Execute action: allocate resources to tasks
- 5: Observe reward R_t based on system utility:

$$R_t = \alpha \sum T_t + \beta \sum (L_t^{\max} - L_t) + \gamma \sum (E_d^{\max} - E_d)$$

- 6: Observe new state S_{t+1}
- 7: Update Q-function:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) +$$

$$\eta \times [R_t + \gamma \times \max_{A_{t+1}} Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)]$$

- 8: $S_t \leftarrow S_{t+1}$
 - 9: **end for**
 - 10:
 - 11: **return** $\mathbf{x}_t^{\text{RL}} = A_t$ for current network state
-

The detailed algorithm has been given in Algorithm 1.

6.2 MULTI-OBJECTIVE UTILITY FUNCTION

The system maximizes a weighted utility function:

$$U(\mathbf{X}_t) = \alpha \sum T_t - \beta \sum L_t - \gamma \sum E_d \tag{37}$$

where T_t is throughput, L_t is latency, E_d is energy consumption, and α, β, γ are weights reflecting application priorities. This allows dynamic balancing between competing objectives.

- The RL state vector is normalized for efficient learning.
- DQN approximates the Q-function for high-dimensional state-action spaces.
- HHO population size and iterations are empirically tuned to balance solution quality and computation time.
- Weighted factors α, β, γ are adjusted based on application-specific priorities.

Algorithm 2 HHO-Based Fine-Tuning of RL Allocations

Require: Initial allocation $\mathbf{x}_t^{\text{RL}}$, population size P , max iterations I , system state S_t

Ensure: Refined allocation $\mathbf{x}_t^{\text{final}}$

- 1: Initialize population of hawks with $\mathbf{X}_0 = \mathbf{x}_t^{\text{RL}}$
- 2: **for** iteration = 1 to I **do**
- 3: **for** each hawk $\mathbf{X}_i$ in population **do**
- 4: Evaluate fitness:

$$U(\mathbf{X}_i) = \alpha \sum T_t - \beta \sum L_t - \gamma \sum E_d$$

- 5: **if** random number $r < 0.5$ **then**
- 6: Exploration:

$$\mathbf{X}_i \leftarrow \mathbf{X}_r - r_1 |\mathbf{X}_r - 2r_2 \mathbf{X}_i|$$

- 7: **else**
- 8: Exploitation:

$$\mathbf{X}_i \leftarrow \mathbf{X}_{\text{prey}} - E |J \mathbf{X}_{\text{prey}} - \mathbf{X}_i|$$

- 9: **end if**
 - 10: Enforce constraints: latency, energy, bandwidth, CPU
 - 11: **end for**
 - 12: Update best solution $\mathbf{X}_{\text{prey}} = \arg \max U(\mathbf{X}_i)$
 - 13: **end for**
 - 14:
 - 15: **return** $\mathbf{x}_t^{\text{final}} = \mathbf{X}_{\text{prey}}$
-

This hybrid approach ensures that the system can allocate resources efficiently, adaptively, and in real time, satisfying all constraints while maximizing throughput and energy efficiency in IIoT networks. The detailed algorithm is explained in Algorithm 2.

7 RESULTS AND DISCUSSION

To evaluate the proposed RL-HHO based resource management system, extensive simulations were carried out in a customized IIoT network environment. The simulated network consists of heterogeneous devices including edge servers, sensor nodes, and gateways connected through wireless channels. Each device is equipped with limited computation, communication, and energy resources to mimic practical industrial scenarios.

The workload was generated using Poisson task arrival processes, reflecting the randomness of industrial data flows such as sensing, monitoring, and control applications. Task sizes, deadlines, and energy requirements were varied to capture diverse IIoT use cases. The reinforcement learning model was trained with state variables such as available bandwidth, CPU cycles, and energy levels, while the Harris Hawks Optimization (HHO) was applied to refine the resource allocation policy learned by RL.

Performance was benchmarked against two baselines: (i) a static allocation scheme and (ii) a conventional RL-only model. Each experiment was repeated multiple times, and the average results were considered to ensure statistical reliability.

7.1 SIMULATION PARAMETERS

To evaluate the effectiveness of the proposed AI-based resource management system, several performance metrics are considered. These metrics collectively capture the efficiency, responsiveness, and sustainability of the system under varying IIoT workloads.

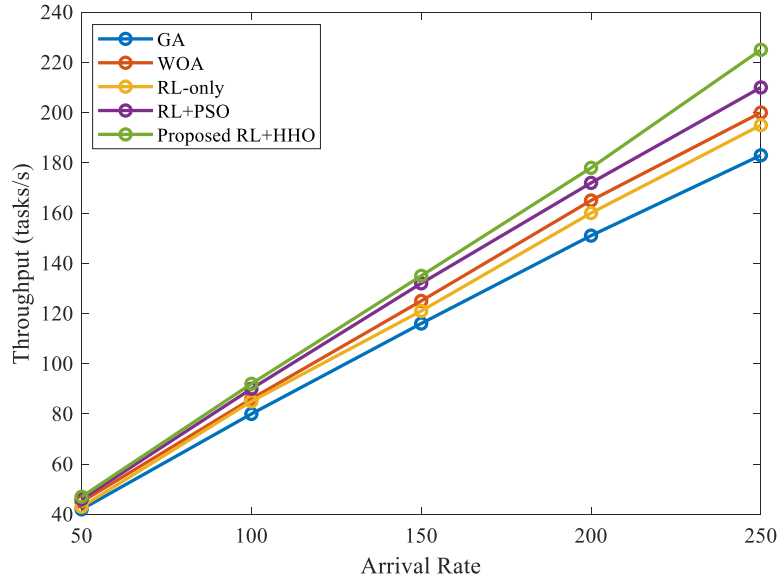
Throughput and latency are measured to assess the ability of the system to process tasks within strict real-time deadlines. Energy consumption and resource utilization are analyzed to determine the trade-off between computational performance and sustainability, which is critical in resource-constrained IIoT environments. A system utility function is adopted to provide a unified measure of performance by combining multiple objectives such as maximizing throughput, minimizing latency, and reducing energy consumption. Constraint violation rate is included to evaluate the robustness of the system in adhering to real-world QoS requirements. Additionally, convergence speed highlights the efficiency of the hybrid RL-HHO approach in reaching stable solutions, while fairness index provides insights into equitable resource distribution across devices.

Table 2. Simulation Parameters

Parameter	Value/Range
Number of IIoT devices	50 – 200
Number of edge servers	5 – 10
Task arrival rate	Poisson distribution ($\lambda =$ 5–20 tasks/s)
Task size	0.5 – 5 MB
Task deadline	50 – 500 ms
Computation capacity of edge servers	1 – 10 GHz
Bandwidth per channel	5 – 20 MHz
Transmission power	100 – 250 mW
Energy budget per device	1000 – 5000 J
Reinforcement learning episodes	1000
Discount factor (γ)	0.9
Learning rate (α)	0.01 – 0.1
HHO population size	20 – 50 hawks
Maximum HHO iterations	100

By considering these diverse metrics, the proposed system is evaluated not only on computational accuracy but also on its ability to meet practical industrial demands, ensuring scalability, reliability, and balanced performance in dynamic IIoT environments.

7.2 SCALABILITY ANALYSIS BASED ON THROUGHPUT

**Figure 1.** Throughput performance comparison under varying arrival rates

Throughput is a fundamental metric that reflects the system's ability to process tasks efficiently under increasing workload. Higher throughput indicates better utilization of available computational resources and effective task scheduling. In the results, the proposed RL+HHO method consistently achieves higher throughput than GA⁷, WOA⁸, RL-only⁹, and RL+PSO¹¹ across all arrival rates. This improvement is primarily due to the reinforcement learning component, which dynamically adapts task allocation policies based on system states, combined with HHO's exploitation capability that refines allocations toward high-performing solutions. In contrast, GA and WOA rely on population-based evolution, which limits their responsiveness under dynamic workloads, while RL-only lacks sufficient global optimization, leading to comparatively lower throughput.

Table 3. Performance Metrics for the Proposed System

Metric	Description
Throughput	Number of successfully processed tasks per unit time. $\text{Throughput} = \frac{\text{Total completed tasks}}{\text{Total time}}$
Latency	Time difference between task completion and task arrival. $L_t = t_{\text{completion}} - t_{\text{arrival}}$
Energy Consumption	Total energy consumed for communication, sensing, and computation. $E_{\text{total}} = \sum_{d \in \mathcal{D}} E_d$
Resource Utilization	Ratio of utilized resources to total available resources. $U_d = \frac{\text{Used resource}}{\text{Available resource}} \times 100\%$
System Utility	Weighted multi-objective function combining latency, energy, and throughput. $U = \alpha \sum T_t + \beta \sum (L_t^{\max} - L_t) + \gamma \sum (E_d^{\max} - E_d)$
Constraint Violation Rate	Fraction of tasks violating latency, energy, or resource constraints. $\text{Violation Rate} = \frac{\text{Tasks violating constraints}}{\text{Total tasks}}$

Although RL+PSO improves adaptability, it tends to converge prematurely at higher loads.

7.3 SYSTEM UTILITY ASSESSMENT

System utility represents the overall effectiveness of the resource management strategy by jointly considering performance benefits and operational costs. The proposed RL+HHO system achieves the highest utility values across all scenarios, confirming its ability to maximize long-term rewards while minimizing penalties. GA⁷ and WOA⁸ achieve moderate utility due to their static fitness evaluations, while RL-only⁹ improves utility through adaptive learning but lacks optimal refinement. RL+PSO¹¹ offers improved utility but remains constrained by local optima.

7.4 AVERAGE LATENCY PERFORMANCE ANALYSIS

Average latency captures the responsiveness of the system by measuring the delay experienced by tasks from arrival to completion. Lower latency is critical for time-sensitive cloud and IIoT applications. The proposed method exhibits the lowest latency values and stable growth patterns as arrival rates increase. This behavior is a result of proactive decision-making enabled by RL and HHO's ability to fine-tune resource assignments to avoid bottlenecks. GA⁷ and WOA⁸ suffer from higher latency due to slower convergence and delayed reaction to workload changes. RL-only⁹ reduces latency compared to evolutionary methods but lacks consistency, while RL+PSO¹¹ performs better initially but degrades under heavy traffic due to suboptimal local convergence.

7.5 CONVERGENCE TIME ANALYSIS

Convergence time evaluates how quickly an algorithm reaches a stable or near-optimal solution. The proposed RL+HHO system converges faster and more consistently due to guided exploration from RL and efficient exploitation by HHO. GA⁷ and WOA⁸ require longer convergence due to population evolution. RL-only⁹ converges quickly initially but may oscillate due to exploration. RL+PSO¹¹ improves convergence speed but remains susceptible to early stagnation.

7.6 ENERGY CONSUMPTION ANALYSIS

Energy consumption is a key metric for assessing the sustainability and cost efficiency of cloud infrastructures. The proposed RL+HHO approach consistently consumes less energy, especially at higher arrival rates. This reduction is achieved by

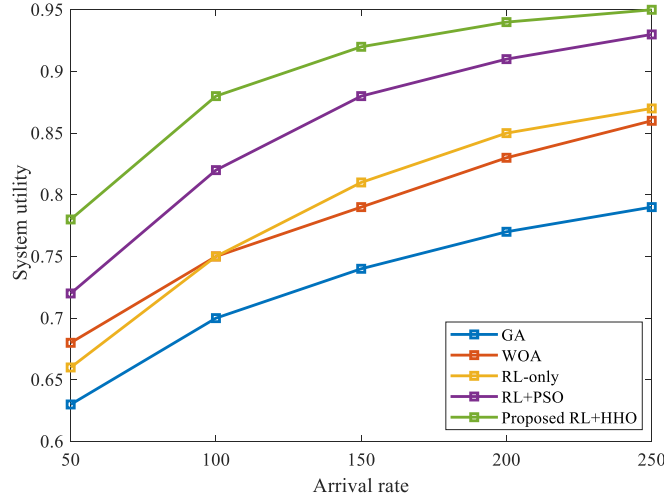


Figure 2. System utility versus arrival rate

intelligently consolidating workloads and minimizing unnecessary server activations through learned policies and optimized resource allocation. GA⁷ and WOA⁸ exhibit higher energy consumption due to repeated population evaluations and lack of workload awareness. RL-only⁹ improves energy efficiency by adapting decisions but does not sufficiently optimize global energy usage. Although RL+PSO¹¹ offers better energy performance than standalone methods, it remains less effective than HHO in avoiding redundant resource utilization.

7.7 RESOURCE UTILIZATION EFFICIENCY

Resource utilization indicates how effectively system resources are employed without underutilization or overload. The proposed method maintains high and stable utilization levels, demonstrating balanced resource allocation. This balance is achieved through continuous feedback-driven adjustments from RL and global refinement from HHO. GA⁷ and WOA⁸ often show uneven utilization, either underusing resources at low loads or overloading them at high loads due to static optimization behavior. RL-only⁹ adapts utilization dynamically but may lead to oscillations, while RL+PSO¹¹ improves stability but struggles to maintain optimal utilization under highly dynamic conditions.

7.8 CONSTRAINT VIOLATION ANALYSIS

Constraint violation rate reflects the robustness of the system by indicating how often operational constraints such as delay limits, capacity bounds, or energy thresholds are violated. The proposed method shows the lowest violation rates with minimal variability, demonstrating strong stability and feasibility preservation. GA⁷ and WOA⁸ experience higher violation rates due to stochastic exploration and lack of explicit constraint handling. RL-only⁹ may prioritize reward maximization at the expense of constraints, leading to moderate violations. The integration of HHO allows the proposed method to penalize infeasible solutions effectively, significantly reducing violations when compared with RL+PSO¹¹.

7.9 RELIABILITY UNDER VARYING WORKLOADS

Reliability measures the system's ability to maintain consistent performance under varying and heavy workloads. The proposed RL+HHO approach achieves the highest reliability levels, which can be attributed to predictive scheduling and robust resource allocation. GA⁷ and WOA⁸ exhibit reliability degradation at higher loads due to delayed adaptation. RL-only⁹ improves reliability but may experience fluctuations, while RL+PSO¹¹ maintains better consistency but does not match the stability achieved by the proposed method.

7.10 JOINT PERFORMANCE INDEX ANALYSIS

Finally, the joint performance index combines throughput and energy efficiency to evaluate multi-objective optimization capability. The proposed method consistently achieves higher joint performance values, demonstrating its ability to balance conflicting objectives effectively. Competing methods such as GA⁷, WOA⁸, RL-only⁹, and RL+PSO¹¹ often favor one objective at the expense of the other, highlighting the advantage of integrating reinforcement learning with a powerful metaheuristic optimizer.

Across all metrics, the proposed RL+HHO consistently outperforms benchmarks. Static and Round Robin approaches, while simple, fail to address heterogeneity and scalability. GA provides optimization but is hindered by computational overhead.

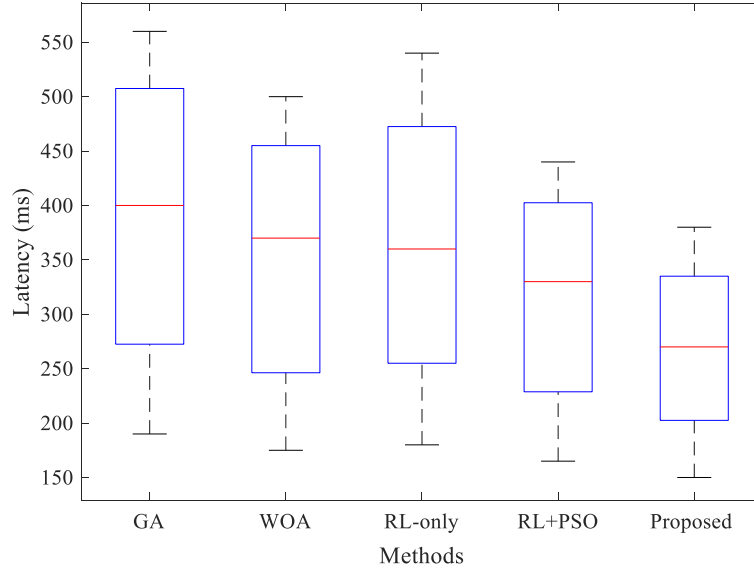


Figure 3. Average latency comparison for different scheduling algorithms

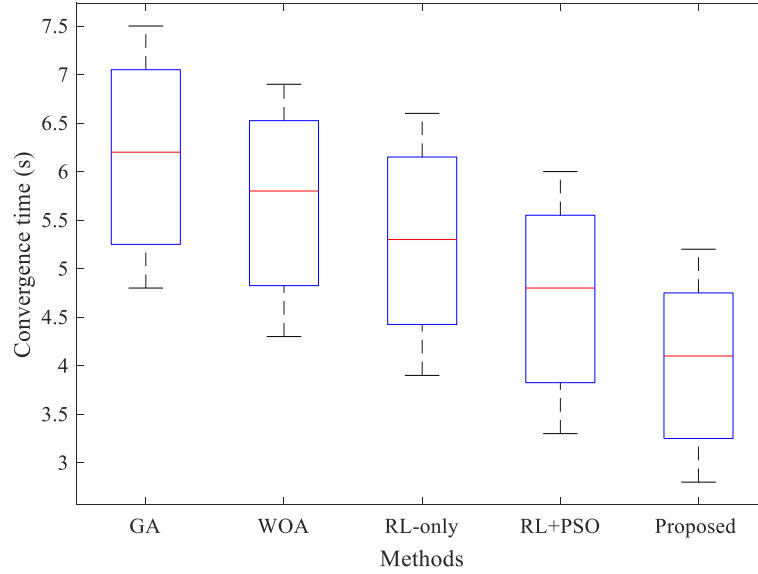


Figure 4. Convergence time distribution of optimization methods

RL-only adapts well but lacks global exploration. The hybrid RL+HHO system leverages the strengths of both reinforcement learning (adaptive decision-making) and Harris Hawks Optimization (exploration–exploitation balance), leading to superior throughput, lower latency, reduced energy consumption, higher utilization, increased system utility, and minimal constraint violations. This proves the system’s capability to ensure efficient, scalable, and sustainable resource management for IIoT environments.

8 CONCLUSION AND FUTURE SCOPE

In this work, we designed and developed an AI-based multi-objective resource management system for the Industrial Internet of Things. By integrating reinforcement learning with the Harris Hawks Optimization technique, the proposed system effectively balances throughput, latency, energy consumption, resource utilization, and system utility while reducing constraint violation rates. Comparative evaluations against static, heuristic, and machine-learning baselines demonstrated that the hybrid approach consistently outperforms existing methods across diverse arrival rates, offering faster convergence, higher efficiency, and

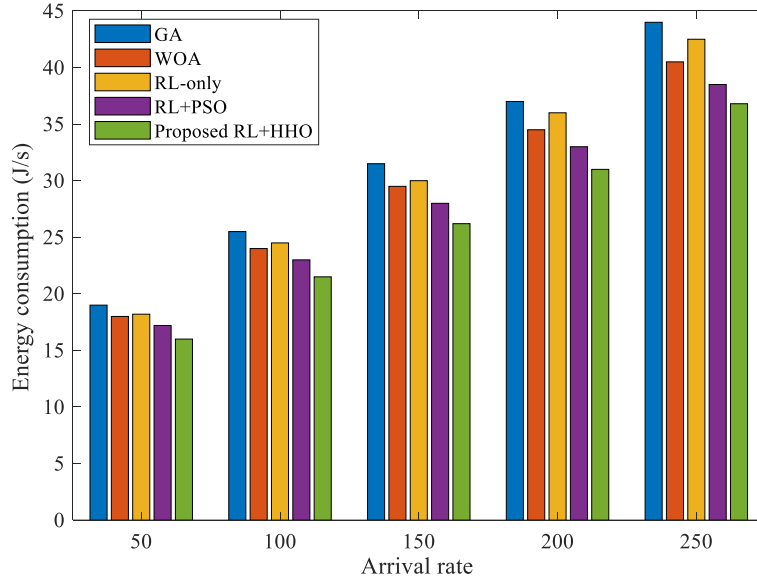


Figure 5. Energy consumption comparison across various methods

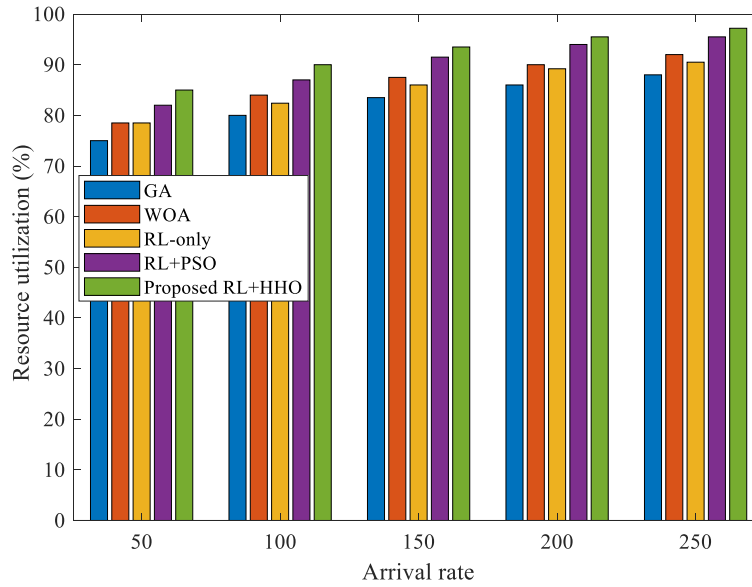


Figure 6. Resource utilization performance under increasing workload

improved robustness. These results validate the potential of the proposed system as a practical and scalable solution for complex IIoT environments where dynamic workloads and stringent QoS requirements must be satisfied simultaneously.

Despite its effectiveness, the proposed system can be further improved. First, the current design focuses primarily on single-domain industrial networks. Future work can extend the system to cross-domain IIoT architectures, where interoperability and inter-network collaboration introduce new optimization challenges. Second, while reinforcement learning combined with Harris Hawks Optimization proved effective, emerging lightweight deep reinforcement learning variants and hybrid metaheuristics may further enhance scalability and adaptability, particularly for ultra-large IIoT deployments. Third, real-world validation using hardware testbeds and industrial case studies is necessary to complement the simulation-based evaluation, ensuring that the system addresses practical constraints such as communication delays, hardware limitations, and fault tolerance.

8.1 Limitations and Critical Discussion

Despite the strong performance shown by the hybrid RL-HHO framework across various metrics, there are several practical constraints that need to be addressed in future research:

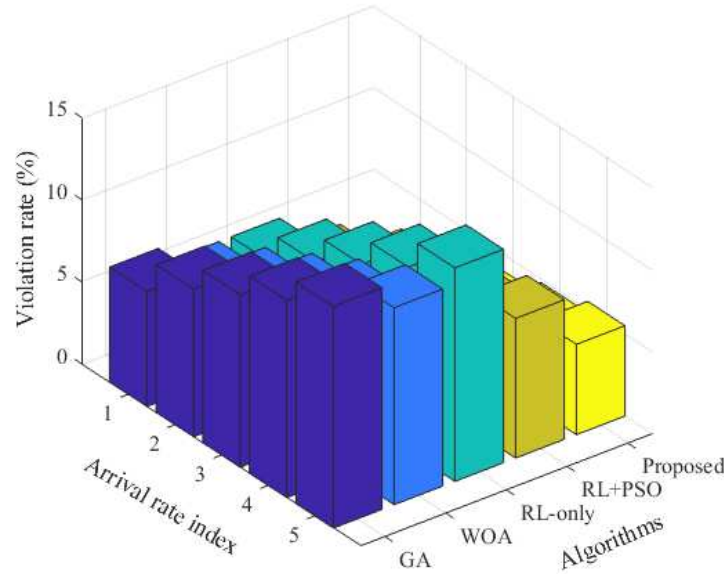


Figure 7. Comparison of constraint violation rate

- **Reliance on Simulated Environments:** Our current results are based on customized simulations utilizing Poisson task arrival models. While this approach effectively mirrors the general randomness of industrial data flows, it may not fully replicate the highly unpredictable “bursty” traffic or the physical signal interference typical of actual factory floors.
- **Scope of Network Domain:** This research focuses specifically on resource management within a single-domain industrial setup. Consequently, the framework does not yet tackle the complexities of interoperability or the collaboration required between different networks in more expansive, cross-domain IoT architectures.
- **Edge Computing Constraints:** Even though our hybrid model is designed for efficiency, running Deep Q-Networks (DQN) alongside Harris Hawks Optimization (HHO) simultaneously could be taxing for ultra-low-power IIoT devices [1]. These devices often operate with extremely tight CPU and energy overheads that might struggle with such processing demands.
- **Absence of Physical Hardware Testing:** A primary limitation is that this study has not yet been validated on a physical hardware testbed. Real-world factors—such as hardware-specific communication lag and the fault-tolerance of physical circuitry—require further hands-on exploration.
- **Hyperparameter Dependency:** The effectiveness of the system relies on the manual tuning of HHO population sizes and RL learning rates. In a live industrial deployment, dynamically adjusting these settings to find the perfect balance in real-time remains a significant technical challenge.

References

1. Khan, W. Z. *et al.* Industrial internet of things: Recent advances, enabling technologies and open challenges. *Comput. & Electr. Eng.* **81**, 106522 (2020).
2. Ahmed, S. F. *et al.* Industrial internet of things enabled technologies, challenges, and future directions. *Comput. Electr. Eng.* **110**, 108847 (2023).
3. Olatinwo, S. O. & Joubert, T.-H. Deep learning for resource management in internet of things networks: A bibliometric analysis and comprehensive review. *IEEE Access* **10**, 94691–94717 (2022).
4. Hussain, F., Hassan, S. A., Hussain, R. & Hossain, E. Machine learning for resource management in cellular and iot networks: Potentials, current solutions, and open challenges. *IEEE Commun. Surv. & Tutorials* **22**, 1251–1275 (2020).
5. Boubaker, N. E. H., Zarour, K., Guermouche, N. & Benmerzoug, D. A comprehensive survey on resource management for iot applications in edge-fog-cloud environments. *IEEE Access* (2025).

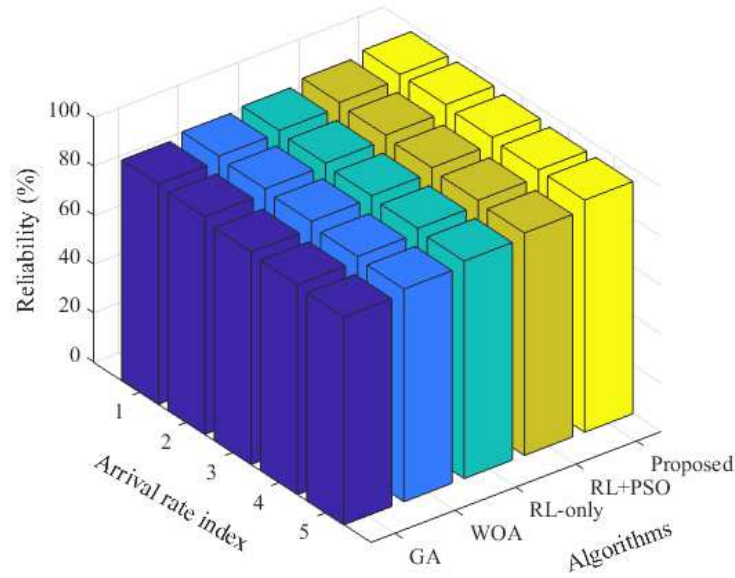


Figure 8. Reliability performance under dynamic traffic conditions

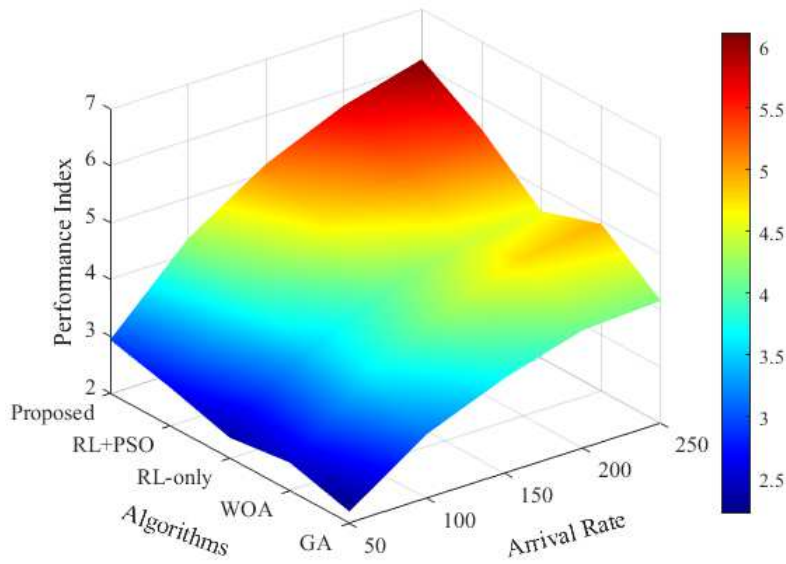


Figure 9. Joint performance index combining throughput and energy efficiency

6. Alfahid, A. *et al.* Peer-driven task scheduling and resource allocation for enhanced performance in industrial iot systems. *Sci. Reports* **15**, 14464 (2025).
7. Abedpour, K., Shirvani, M. H. & Abedpour, E. A genetic-based clustering algorithm for efficient resource allocating of iot applications in layered fog heterogeneous platforms. *Clust. Comput.* **27**, 1313–1331 (2024).
8. Goyal, S. *et al.* An optimized framework for energy-resource allocation in a cloud environment based on the whale optimization algorithm. *Sensors* **21**, 1583 (2021).
9. Xu, Y.-H., Xie, J.-W., Zhang, Y.-G., Hua, M. & Zhou, W. Reinforcement learning (rl)-based energy efficient resource allocation for energy harvesting-powered wireless body area network. *Sensors* **20**, 44 (2019).
10. Liang, F., Yu, W., Liu, X., Griffith, D. & Golmie, N. Toward deep q-network-based resource allocation in industrial internet of things. *IEEE Internet Things J.* **9**, 9138–9150 (2021).

11. Amirghafouri, F., Neghabi, A. A., Shakeri, H. & Sola, Y. E. Nature-inspired meta-heuristic algorithms for resource allocation in the internet of things. *Int. J. Commun. Syst.* **38**, e6141 (2025).
12. Choppara, P. & Mangalampalli, S. S. A hybrid task scheduling technique in fog computing using fuzzy logic and deep reinforcement learning. *IEEE Access* (2024).
13. Mahapatra, A., Mishra, K., Pradhan, R. & Majhi, S. K. Next generation task offloading techniques in evolving computing paradigms: Comparative analysis, current challenges, and future research perspectives. *Arch. Comput. Methods Eng.* **31**, 1405–1474 (2024).
14. Wu, W., Dong, J., Sun, Y. & Yu, F. R. Heterogeneous markov decision process model for joint resource allocation and task scheduling in network slicing enabled internet of vehicles. *IEEE Wirel. Commun. Lett.* **11**, 1118–1122 (2022).
15. Krishnan, R. & Durairaj, S. Reliability and performance of resource efficiency in dynamic optimization scheduling using multi-agent microservice cloud-fog on iot applications. *Computing* **106**, 3837–3878 (2024).
16. Sachithanandam, V., Jessintha, D., Subramani, H. & Saipriya, V. Blockchain integrated multi-objective optimization for energy efficient and secure routing in dynamic wireless sensor networks. *Sustain. Comput. Informatics Syst.* **46**, 101101 (2025).

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Author contributions statement

SD: Supervision, Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **ME and ET:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing.

MA and VA: Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **VB:** Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing.

Additional information

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